Nursing Practice Guideline

Non-invasive Positive Pressure Ventilation (NIPPV) and Continuous Positive Airways Pressure (CPAP)

Reference #: NPG69

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	Nursing

Introduction

This guideline outlines the procedures and responsibilities of individuals and the health service organisation in relation to Non-invasive Positive Pressure Ventilation (NIPPV) and Continuous Positive Airways Pressure (CPAP). It ensures the delivery of a uniform, integrated multidisciplinary approach with management that is consistent with best practice.

Risk Statement

Non-compliance with this guideline will:

Breach legislative requirements	☒	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☐
Breach SCGH Mission & Values	☒	Staff Safety	☒
Other:	☐		

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1. Principles

Observe the 5 moments for hand hygiene and wear appropriate personal protective equipment when managing NIPPV, CPAP or manipulating the circuit; follow the risk assessment outlined in SCGOPHCG Infection Prevention and Control Manual Aseptic Technique Policy.

All NIPPV circuits must be changed every 7 days or PRN if visibly soiled.[8]
Mask to be wiped out daily with detergent and water and dried thoroughly.
Patients own equipment to be washed with soapy water and dried thoroughly every seven days.

2. Non-Invasive Positive Pressure Ventilation (NIPPV)- SCGH Only

2.1 Medical Requirements

NIPPV is prescribed by a Consultant or Registrar, in consultation with the Respiratory/Pulmonary Physiology team where appropriate.

For patients commenced on NIPPV for respiratory failure consider a referral to the on-call respiratory team for ongoing review and management of NIPPV.

NIPPV is indicated in the following clinical scenarios:

- Acute exacerbations of chronic obstructive pulmonary disease (COPD) [1-3],
- Hypercapnoeic respiratory failure,
- Weaning from tracheal intubation [3, 4],
- Palliation of breathlessness in progressive neuromuscular diseases[2, 5],
- Management of chronic respiratory failure (sub-acute),
- Sleep hypoventilation (neuromuscular and chest wall restrictive disorders and obesity hypoventilation syndrome)[3].

Selected cases of:

- acute pulmonary oedema (where continuous positive airways pressure (CPAP) is inadequate)[3, 6],
- pneumonia (e.g. immunocompromised patients, COPD) [3, 7],
- cystic fibrosis (as a bridge to transplantation)[2]

Medical staff is required to:

- Document a clear plan for the parameters indicating escalation to intubation or decision not to intubate in the event of NIPPV failure.[3]
- Explain NIPPV therapy to the patient and obtain verbal consent where possible.
- Prescribe Inspiratory Positive Airway Pressure - IPAP, Expiratory Positive Airway Pressure - EPAP, target oxygen parameters:

IPAP	Inspiratory Positive Airway Pressure
EPAP	Expiratory Positive Airway Pressure
PEEP	Positive End Expiratory Pressure
PS	Positive Pressure Support
CPAP	Continuous Positive Airway Pressure

Summary of ventilator terms

IPAP – EPAP = PS
PS + PEEP = IPAP
PEEP = CPAP

- Document the NIPPV prescription, humidification, flow rate of supplemental oxygen and target SpO_2 on the HFNO/CPAP/NIV Management Chart MR876.

- Oxygen is capped at **30L/minute** via Astral© Non-Invasive Ventilator (G54 ward device).[11]
- Reassess the patient 1-2 hours after commencement of NIPPV or earlier if clinically indicated. Document any changes in the NIPPV prescription on the NIV prescription chart.
- Review and re-prescribe NIPPV daily.
- Discontinue NIPPV therapy.

Arterial (ABG) or capillary CBG) blood gases are required:

- Prior to commencement of NIPPV,
- 2, 6, and 24 hours after initiation of therapy,
- If ABG or CBG analysis is unavailable or not feasible, venous blood gas (VBG) analysis can be used with good correlation between pH, HCO_3 and base excess; consider acute hypercapnoeic respiratory failure with venous pH of <7.32 and PvCO2 >50mmHg [12,13]
- Any patient with an SpO_2 < 80% requires an immediate ABG [13]

2.2 Safety Information

Endotracheal intubation should be considered in the following circumstances:

- Cognitive impairment - reduced consciousness, confusion or agitation,
- Medically unstable - imminent cardiac or respiratory arrest, life threatening hypoxaemia or severe acidosis,
- Untreated pneumothorax,
- Inability to protect airway or vomiting,
- Bowel obstruction,
- Oro-facial abnormalities interfering with mask fit – e.g. burns, trauma, and recent facial/upper airway surgery.

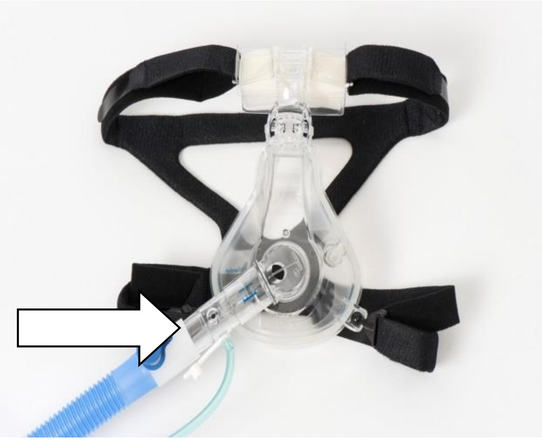
2.3 Nursing Management

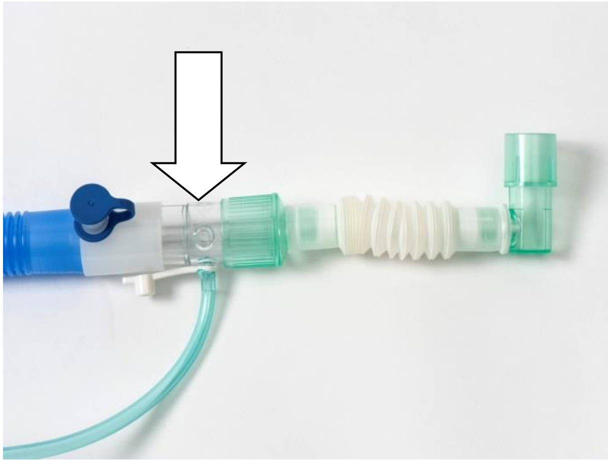
- Referral to the Intensive Care Unit (ICU) or Respiratory Special Care Unit (RSCU) area should be considered.
- Patients receiving NIPPV for acute respiratory failure or acute pulmonary oedema outside Intensive Care Unit (ICU), Coronary Care Unit (CCU), the respiratory ward, GHDU, or Neurosurgery HDU may require a nurse special. Ward CNS / Afterhours CNS to review need for special. [3] (See section 3 for management of patients on home care NIPPV/CPAP).

The nurse will:

- Reassure and encourage the patient receiving NIPPV to minimise anxiety and improve compliance.
- Sit the patient in an upright or semi-recumbent position to assist with ventilation. Those requiring supine positioning for investigations should be closely monitored.[3]
- Ensure the mask is comfortable. Apply the mask to achieve the best seal to minimise air leaks, and the least tension in the straps to minimise the risk of skin erosion. [3, 7]
- Ensure patient has means of communication whilst mask insitu (call bell, paper and pen)[7].
- Ensure that air is not blowing directly into the patient's eyes and provide regular eye care. [7]
- Provide regular facial pressure area care. Duoderm, gecko© or other padded dressings may be applied. [2, 3, 7]
- Ensure that the CO_2 exhalation port remains patent. (See Figure 1)

Figure 1. Examples of CO_2 exhalation ports





Note: ICU ventilators may not require a CO2 exhalation valve – refer to ICU departmental Guideline.

- Observe for abdominal distension due to aerophagia (gastric air).[3]
- Document any significant change in the patient's status and escalate care as per SCGHOPHCG Recognising and Responding to Acute Deterioration hospital policy

Monitoring of patients receiving NIPPV

- All patients receiving NIPPV require regular observations of the following:
  - Core physiological observations (SpO_2, pulse rate, respiratory rate, blood pressure, level of consciousness)[3, 7],
  - Patient comfort and use of accessory muscles of respiration[3, 7],
  - Coordination of respiratory effort with the ventilator; motion of the chest wall [2, 3, 7],
  - Ventilator settings - IPAP and EPAP, leak, tidal volumes (Vt),
  - Flow rate of supplemental oxygen L/min and FiO2 (oxygen capped at 30L/minute via Astral NIV machine).[11]
  - Fluid Balance Chart (FBC) if evidence of heart failure.
- In patients commencing NIPPV for acute respiratory failure or acute pulmonary oedema, these observations should be recorded every 15 minutes for the first hour or until stable. More frequent observations may be required for highly unstable patients.[3]
- Once stable, these observations are recorded hourly while the patient is on NIPPV. Blood pressure is monitored hourly for the first 2 hours and then 4 hourly if stable.[3]
- When not on NIPPV observations are recorded as per Observation and Response Chart (ORC) escalation protocol.

- Patients who show benefit from NIPPV in the first few hours should be ventilated as much as possible during the first 24 hours. However, NIPPV can be interrupted while the patient is awake for medications, physiotherapy, meals, etc.[1] SpO_2 target parameters should be documented on the Non-Invasive Ventilation (NIV) Prescription Chart (MR 876).
- Administer bronchodilators as prescribed[9]
- Referral to Physiotherapist.[3]

Treatment Failure

The following findings indicate a need to review the diagnosis and management.[2,3,7]

- Worsening level of consciousness or increase in agitation
- Worsening respiratory distress or failure to alleviate symptoms
- Worsening gas exchange - fall in SpO_2, increase in O_2 requirements or deterioration of ABG/CBG tensions
- Deteriorating core observations - increase in pulse or respiratory rate
- Intolerance of or failure of coordination with ventilator

If appropriate, endotracheal intubation should be considered, and the patient discussed with the Intensive Care Unit.

Discontinuation of NIPPV therapy

NIPPV therapy may be discontinued when: [2, 3, 7]

- NIPPV no longer required because of clinical improvement
- NIPPV not tolerated by patient
- Complication of NIPPV eg: pneumothorax
- Endotracheal intubation required because of NIPPV treatment failure
- Patient elects to discontinue therapy

The decision to discontinue NIPPV or therapy is made by the Registrar or Consultant after appropriate discussion with the patient. This must be documented.

The patient may require supplemental oxygen, titrated to maintain target SpO2. Monitor for signs of deterioration in patient's clinical condition such as:

- Worsening level of consciousness,
- Increase in respiratory distress,
- Worsening of core observations – increase in pulse or respiratory rate,
- Worsening gas exchange - fall in SpO2, increase in O2 requirements or deterioration of ABG tensions.

2.4 Procedural Information

NIPPV equipment for transfers is available for loan from the Respiratory Ward, G54. Call Ward G54 with patient details prior to sending HSA to collect equipment.

Requirements

- NIV machine
- Bacteriostatic filter
- Full face mask and head strap
- Oxygen tubing
- Adult Breathing Circuit Inspiratory Heated with MR290 Autofeed Chamber
- Sterile water 1L bag
- Dedicated pulse oximeter (not available from G54)
- HFNO/CPAP/NIV management chart (MR 876)
- Suction equipment

Figure 2: Set up of NIPPV

Procedure

Application of NIPPV

1. Prior to applying NIPPV ensure the equipment is correctly assembled and working and oxygen is connected to the circuit. Supplemental oxygen may be required when NIPPV is used in acute respiratory failure or pulmonary oedema. Ensure the CO_2 exhalation port is patent. Connect the NIPPV ± humidifier to mains power (Astral© NIV machine has in built back-up battery supply for transport) and turn the ventilator ± humidifier on. In an emergency situation NIPPV may be commenced without humidification.
2. Size the patient for the appropriate mask:[3]
    - Full-face mask – the mask should comfortably cover the nose and mouth, it should not cover the eyes or rest below the chin.
    - Nasal mask – it should not be as large as to cover any part of the mouth, or so small as to block the nares.
    - Total Face mask – the mask seals around the perimeter of the face, it covers the eyes, nose and mouth.
3. Ensure that the mask is placed to achieve the best fit and seal with the least tension on the straps, ensuring that straps do not cover the earlobes. (Figure 2)[3]
4. After fitting the mask check for air leak at interface and ensure air is not blowing directly into the patient's eyes. [3]
5. Once mask is fitted remain at the patient side until they are comfortable and tolerating NIPPV.[3]

Removal of NIPPV

1. Post discontinuation observations as per Observation and Response Chart (ORC).
2. Discard bacteriostatic filter, oxygen tubing, circuit, humidification chamber and sterile water.
3. Place all other equipment into a plastic bag (labelled dirty equipment) and return to the respiratory ward with the mobile NIPPV unit. The respiratory ward will send this equipment to Sterilising Services Department for reprocessing.

3. Continuous Positive Airway Pressure (CPAP)

3.1 Medical Requirements

CPAP is prescribed by a consultant or registrar in consultation with the Respiratory/Pulmonary Physiology/Cardiology team where appropriate.

CPAP is indicated for use in the following scenarios: [2, 3, 6, 7]

- Acute Pulmonary Oedema
- Obstructive Sleep Apnoea
- Sleep hypoventilation (neuromuscular and chest wall restrictive disorders and obesity hypoventilation syndrome)

A doctor is required to:

- Explain CPAP therapy to the patient and obtain verbal consent where possible.
- Prescribe and document settings for CPAP
- Reassess the patient 1-2 hours after commencement of CPAP and document any changes in the CPAP prescription on the NIPPV prescription chart.
- Review and re-prescribe CPAP therapy daily for acute conditions. For patients who utilise CPAP therapy as home treatment an initial prescription is required and reviewed only if clinical condition changes/deteriorates.
- Discontinue CPAP therapy.

3.2 Nursing Management

- Patients receiving CPAP for **acute respiratory failure** or **acute pulmonary oedema** outside ICU, CCU, Respiratory ward or Neurosurgical HDU may require a nurse special. Ward CNS / Afterhours CNS to review need for special.[3]
- Patients receiving CPAP for **chronic obstructive sleep apnoea <u>may not</u>** require a nurse special or high dependency unit admission. Ward CNS/Afterhours CNS to review. (See section 3 for management of patients on home care NIPPV/CPAP)
- Patients receiving CPAP therapy at home should be encouraged to bring their own equipment to hospital.

The nurse will:

- Reassure and encourage the patient receiving CPAP to minimise anxiety and improve compliance.
- Sit the patient in an upright or semi-recumbent position to assist with ventilation. Those requiring supine positioning for investigations should be closely monitored.[3]
- Ensure the mask is comfortable. Apply the mask to achieve the best seal to minimise air leaks, and the least tension in the straps to minimise the risk of skin erosion. [3, 7]
- Ensure that air is not blowing directly into the patient's eyes and provide regular eye care. [3, 7]
- Provide regular facial pressure area care. Duoderm or other padded dressings may be applied.[3]
- Ensure that the CO_2 exhalation port remains patent.
- Observe for abdominal distension due to aerophagia (gastric air). [3, 7]
- Document any significant change in the patient's status and escalate as per Recognising and Responding to Acute Deterioration policy

Monitoring of patients receiving CPAP

All patients receiving CPAP require regular observations of the following:

- Core observations (SpO_2, pulse rate, respiratory rate, blood pressure, level of consciousness),
- Patient comfort and use of accessory muscles of respiration,
- Coordination of respiratory effort with the ventilator; motion of the chest wall,
- Flow rate of supplemental oxygen,
- Fluid Balance Chart (FBC) if evidence of heart failure.
- CPAP observations are recorded on the HFNO/CPAP/NIV management chart. All other observations should be recorded on the ORC.

For patients commencing CPAP for acute pulmonary oedema, these observations should be recorded every 15 minutes for the first hour or until stable. More frequent observations may be required for highly unstable patients, once stable; these observations are recorded at least hourly while the patient is on CPAP. Blood pressure is monitored at least hourly for the first 4 hours and then 4 hourly if stable.

- When off CPAP observations are recorded as per ORC escalation protocol.[3]

3.3 Procedural Information

Requirements

- Mobile NIV unit
- Bacteriostatic filter
- Full face mask and head strap
- Oxygen tubing
- Adult Breathing Circuit Inspiratory Heated with MR290 Autofeed Chamber
- Sterile water 1L bag
- Pulse oximeter
- NIV prescription and observation chart
- Suction equipment

Procedure

Application of CPAP

1. Prior to applying CPAP ensure the equipment is correctly assembled and working and oxygen is connected to the ventilator. Supplemental oxygen may be required when CPAP is used in acute pulmonary oedema. Ensure the CO_2 exhalation port is patent. Connect the CPAP ± humidifier to mains power (or ensure enough battery backup supply for transport) and turn the ventilator ± humidifier on.
2. Size the patient for the appropriate mask:[10]
    - Full-face mask – the mask should comfortably cover the nose and mouth, it should not cover the eyes or rest below the chin.
    - Nasal mask – It should not be as large as to cover any part of the mouth, or so small as to block the nares.
    - Total Face mask – The mask seals around the perimeter of the face, it covers the eyes, nose and mouth.
3. Ensure that the mask is placed to achieve the best fit and seal with the least tension on the straps, ensuring that straps do not cover the earlobes. (Figure 2: page 6)
4. After fitting the mask check for the absence of air leak at interface and ensure air is not blowing directly into the patient's eyes.
5. Once mask is fitted remain with the patient until they are comfortable and tolerating CPAP.

Removal of CPAP

1. CPAP therapy should be discontinued only on the instruction of the prescribing medical officer.
2. Post discontinuation observations as per ORC
3. Discard bacteriostatic filter, oxygen tubing, circuit, humidification reservoir and sterile water.
4. Place all other equipment into a plastic bag (labelled dirty equipment) and return to the respiratory ward with the mobile NIPPV unit. The respiratory ward will send this equipment to Sterilising Services Department for reprocessing.

4. Use of Home NIPPV or CPAP Treatment in Hospital

NIPPV or CPAP is a commonly prescribed home treatment for chronic conditions such as:

- Obstructive sleep apnoea[3]
- Sleep hypoventilation (neuromuscular and chest wall restrictive disorders and obesity hypoventilation syndrome)[3, 6]
- Chronic respiratory failure related to COPD.[1]

This patient group should be encouraged to continue usual use of NIPPV or CPAP using own equipment (machine, mask, tubing) whilst receiving inpatient treatment.

4.1 Medical Requirements

Medical team should review patients at admission for appropriateness of continued use of home treatment and document NIPPV/CPAP plan in medical record.

Patients will require ongoing review of NIPPV/CPAP treatment at condition change or deterioration of clinical condition.

4.2 Safety Information

It is preferable that patients' own CPAP/NIPPV equipment should receive an electrical safety check before use. Complete an EMPAC for electrical safety check in working hours.

Patients with deteriorating core physiological observations whilst using own NIPPV or CPAP may require increased observation as per Section 1: Non Invasive Ventilation (NIPPV) or Section 2: Continuous Positive Airway Pressure (CPAP) of this guideline.

4.3 Nursing Management

- Encourage usual use of NIPPV or CPAP for sleep.
- At least 4/24 core physiological observations (SpO2, pulse rate, respiratory rate, blood pressure, level of consciousness, flow rate of oxygen)
- Deterioration of core physiological observations; trigger review as per ORC escalation protocol.

Related Documents

Legislation:

- Health Practitioner Regulation National Law (WA) Act 2010
- Carer's Recognition Act 2004 (WA).
- Work Health and Safety Act 2020
- Nursing and Midwifery Board of Australia (NMBA) Code of Professional Conduct for Nurses March 2018

Policy:

- NMHS Consent to Treatment Policy
- NMHS Clinical Documentation Policy
- SCGH Patient Identification and Procedure Matching Policy
- SCGOPHCG Infection Prevention and Control Manual - Aseptic Technique Policy
- SCGOPHCG Recognising and Responding to Clinical Deterioration Hospital Policy
- SCGOPHCG Clinical Handover- Communicating for Safety Hospital Policy

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